

Live Honeypot SOC Lab: Catching Real Attackers with Azure and Sentinel

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Detection Rules at a Glance

Rule	Tactic	Technique	Severity
Brute Force Detection	Credential Access	T1110	Medium
Suspicious Process After Login	Execution	T1059	Medium
New Local User Account Created	Persistence	T1136.001	Medium
Security Event Log Cleared	Defense Evasion	T1070.001	High

1. What This Lab Is About

This lab is a basic SOC build, not just a config walkthrough. I created a honeypot in Azure: a Windows VM deliberately exposed to the public internet with its firewalls turned off, so that real attackers on the internet would find it and try to break in. I forwarded its logs into a central repository, wrote detection rules to catch specific attack behaviors, and built a map to visualize where the attacks were actually coming from.

The point of this lab isn't really the honeypot itself. It's to see, first-hand, how fast and how often a system gets attacked the moment it's exposed to the internet with no protection.

Every attack in this lab is real and came from the live internet, not simulated.

The route I took:

```
Create Resource Group
→ Create Virtual Network
→ Create Virtual Machine
→ Edit Network Security Group
→ Configure new inbound rule
→ Log in to the Windows VM
→ Turn off the firewall in the Windows VM
→ Create a Log Analytics Workspace
→ Connect the Log Analytics Workspace to Sentinel
→ Install and configure the Azure Monitor Agent
→ Configure detection rules
→ Upload geolocation data
→ Create the attacker map
```

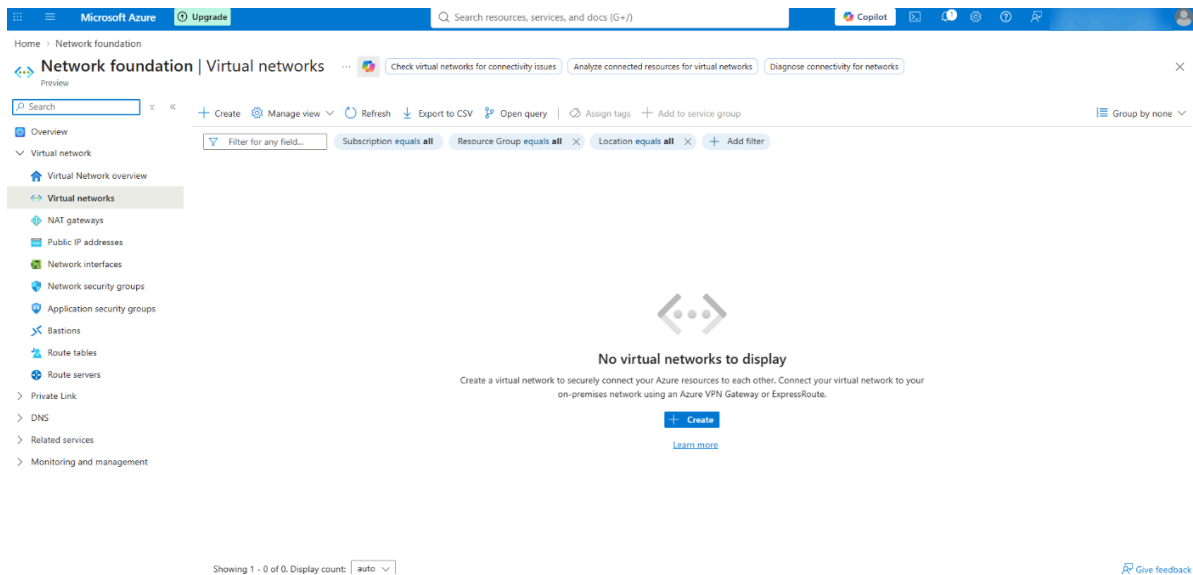
2. Building the Environment

Resource Group: Sentinel-RG

Everything in this lab lives inside one resource group, so the VM, virtual network, and every other resource created for this lab stay grouped together.

Virtual Network

Next I created a virtual network. This gives the honeypot VM a network environment to sit inside and communicate through. After creating it, I checked the resource group to confirm it showed up there, which it did.



Microsoft Azure Upgrade Search resources, services, and docs (G+/I) Copilot

Home > Network foundation | Virtual networks

Network foundation | Virtual networks

Check virtual networks for connectivity issues Analyze connected resources for virtual networks Diagnose connectivity for networks

Search

Create Manage view Refresh Export to CSV Open query Assign tags Add to service group

Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter

Virtual network overview

Virtual networks

NAT gateways

Public IP addresses

Network interfaces

Network security groups

Application security groups

Bastions

Route tables

Route servers

Private Link

DNS

Related services

Monitoring and management

No virtual networks to display

Create a virtual network to securely connect your Azure resources to each other. Connect your virtual network to your on-premises network using an Azure VPN Gateway or ExpressRoute.

+ Create

Learn more

Showing 1 - 0 of 0. Display count: auto Give feedback

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Home > Network foundation | Virtual networks

Create virtual network

Basics Security Address space Tags Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation.

Learn more.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * Sentinel-RG

Create new

Instance details

Virtual network name * VNet-Soc-Lab

Region * (US) East US

Deploy to an Azure Extended Zone

Previous Next Review + create Give feedback

Creating the virtual network inside Sentinel-RG

Virtual Machine

This is the actual honeypot: the VM that's meant to get attacked. Creating it also automatically created a few supporting resources inside the resource group:

- A public IP address, so the VM is reachable from the internet
- A network security group (NSG), which acts as the firewall sitting in front of the VM
- A network interface, which works like the VM's virtual ethernet port
- A disk, for the VM's storage

Microsoft Azure | Upgrade

Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups

Resource Manager | Resource groups

Search

Resource Manager

All resources

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Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Showing 1 - 2 of 2. Display count: auto

Sentinel-RG

Resource group

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Deployments

Security

Deployment stacks

Policies

Properties

Locks

Cost Management

Monitoring

Automation

Help

Essentials

Resources

Recommendations

Filter for any field...

Type equals all

Location equals all

Add filter

Name	Type	Location
SecurityInsights(sentinel-workspace)	Solution	East US
sentinel-workspace	Log Analytics workspace	East US
Vnet-Soc-Lab	Virtual network	East US

Showing 1 - 3 of 3. Display count: auto

Give feedback

Sentinel-RG showing the virtual network and Log Analytics workspace

Microsoft Azure | Upgrade

Search resources, services, and docs (G+)

Home

CreateVirtualMachine-LIFE-NET-EAST-1 | Overview

Deployment

Search

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : CreateVirtualMachine-LIFE-NET-EAST-1

Subscription : Azure subscription 1

Resource group : Sentinel-RG

Start time : 8/31/2026, 5:36:37 AM

Correlation ID : 3f54d5e1-9028-4ed6-961c-787fa8dc2f95

Deployment details

Next steps

Set up auto-shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource

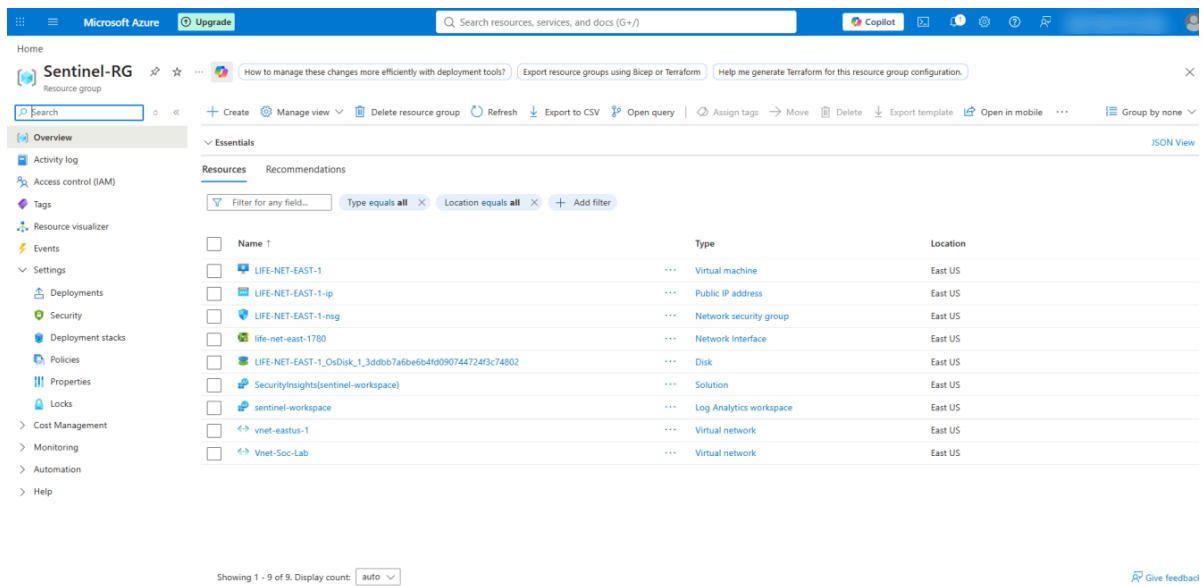
Create another VM

Scale out your VM

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >



VM deployment complete, with all supporting resources created in Sentinel-RG

3. Opening the Firewall (On Purpose)

The network security group is what sits between the VM and the public internet. **Inbound rules** control what traffic is allowed *into* the virtual network from the internet, and **outbound rules** control what's allowed *out*.

To make the honeypot actually attractive to attackers, I deleted the default RDP-only inbound rule and replaced it with my own rule, named `Danger_AllowAnyInbound`, which allows any type of traffic in. In the real world this would be a serious misconfiguration, but for this lab it's the entire point: the VM needs to look wide open for anything to find and attack it.

I set the rule's priority to 100. In Azure NSG rules, a lower priority number means the rule is evaluated sooner. Since there were only a few rules total with plenty of room between their priority numbers, 100 was fine here.

Azure threw several warnings after this change, since I was allowing any traffic inbound. That's expected and not something to panic over, it's Azure correctly flagging that this is an unusual and risky configuration.

Home > Sentinel-RG

LIFE-NET-EAST-1-nsg Network security group

Search [] Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Resource group (magnifying glass): Sentinel-RG

Location: East US

Subscription (magnifying glass): Azure subscription 1

Subscription ID: []

Tags (edit): Add tags

Custom security rules: 1 inbound, 0 outbound

Associated with: 0 subnets, 1 network interfaces

Filter by name [] Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
300	RDP	3389	Tcp	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound Security Rules						
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

Home > LIFE-NET-EAST-1-nsg

LIFE-NET-EAST-1-nsg | Inbound security rules

Search [] Add Hide default rules Refresh Delete Give feedback

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Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

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Automation

Help

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name [] Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Add inbound security rule

LIFE-NET-EAST-1-nsg

Source: Any

Source port ranges: *

Destination: Any

Service: Custom

Destination port ranges: *

Protocol: Any

Action: Allow

Priority: 100

Name: DANGER_AllowsAnyInbound

Add Cancel

Inbound security rules, including the new Danger_AllowAnyInbound rule

Home > LIFE-NET-EAST-1-nsg

LIFE-NET-EAST-1-nsg | Inbound security rules

Search [] Add Hide default rules Refresh Delete Give feedback

Overview

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Diagnose and solve problems

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Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name [] Port == all Protocol == all Source == all Destination == all Action == all

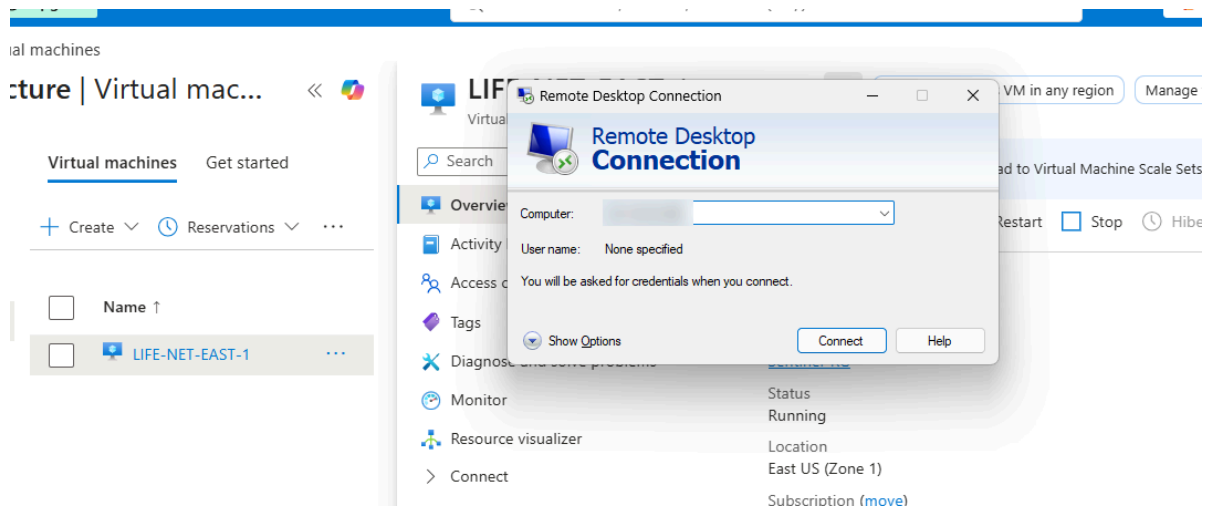
Priority	Name	Port	Protocol	Source	Destination	Action
100	DANGER_AllowsAnyInbound	Any	Any	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Confirming the new inbound rule is active

4. Preparing the VM as a Honeypot

To make the VM vulnerable at the OS level too, not just the network level, I needed to log in and disable its internal Windows Firewall.

1. Copied the VM's public IP address from the Azure portal.
2. Opened Remote Desktop Connection on my local machine and connected using that IP address, along with the username and password set when the VM was created.
3. Once logged in, opened Windows Defender Firewall, turned it off for every network profile tab, and clicked Apply.



Connecting to the VM via Remote Desktop

Windows Security

Enter your credentials

These credentials will be used to connect to [redacted]

User name

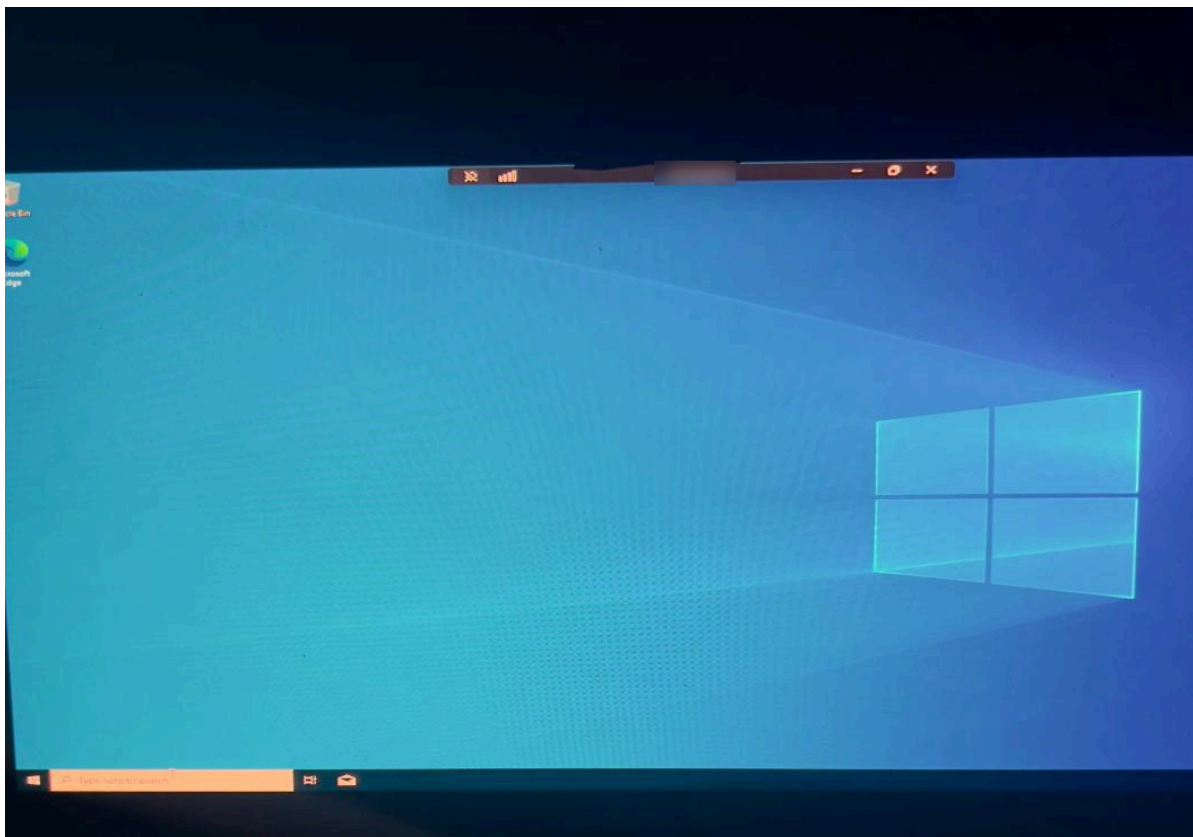
azureuser

Password

[redacted]

☐ Remember me

More choices



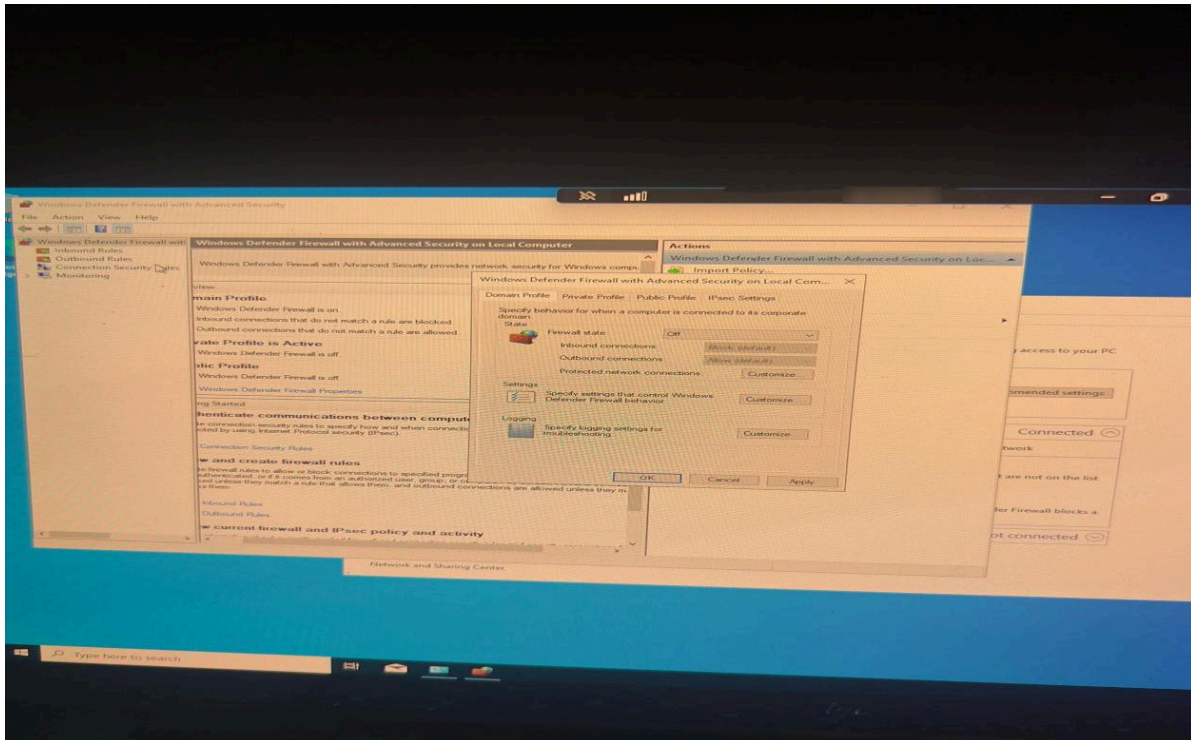
Logged into the honeypot VM's desktop

Confirming the VM was actually reachable from the internet

To check that the VM would actually be visible to attackers, I pinged it from my own computer:

```
ping -l 64 <VM_PUBLIC_IP>
```

-l 64 sets the ping packet size to 64 bytes, and the IP address is the VM's public IP copied from Azure. Getting a response back confirmed the VM was reachable, which meant that if I could reach it, attackers scanning the internet could too.



```
Command Prompt
C:\Users\User>ping -l 64 [redacted]

Pinging [redacted] with 64 bytes of data:
Reply from [redacted]: bytes=64 time=262ms TTL=107
Reply from [redacted]: bytes=64 time=279ms TTL=107
Reply from [redacted]: bytes=64 time=399ms TTL=107
Reply from [redacted]: bytes=64 time=204ms TTL=107

Ping statistics for [redacted]:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 204ms, Maximum = 399ms, Average = 286ms

C:\Users\User>
```

Successful ping response from the exposed VM

5. Setting Up Log Collection

With the VM exposed, I needed a way to actually capture what was happening to it.

I created a Log Analytics Workspace, which acts as the log repository, and connected it to Microsoft Sentinel so I could view and query the logs. It's technically possible to view security events directly in the VM's own Windows Event Viewer, but Sentinel is far easier to work with, especially for querying and filtering across a large volume of events.

Home > Log Analytics workspaces

Create Log Analytics workspace

Basics Tags Review + Create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

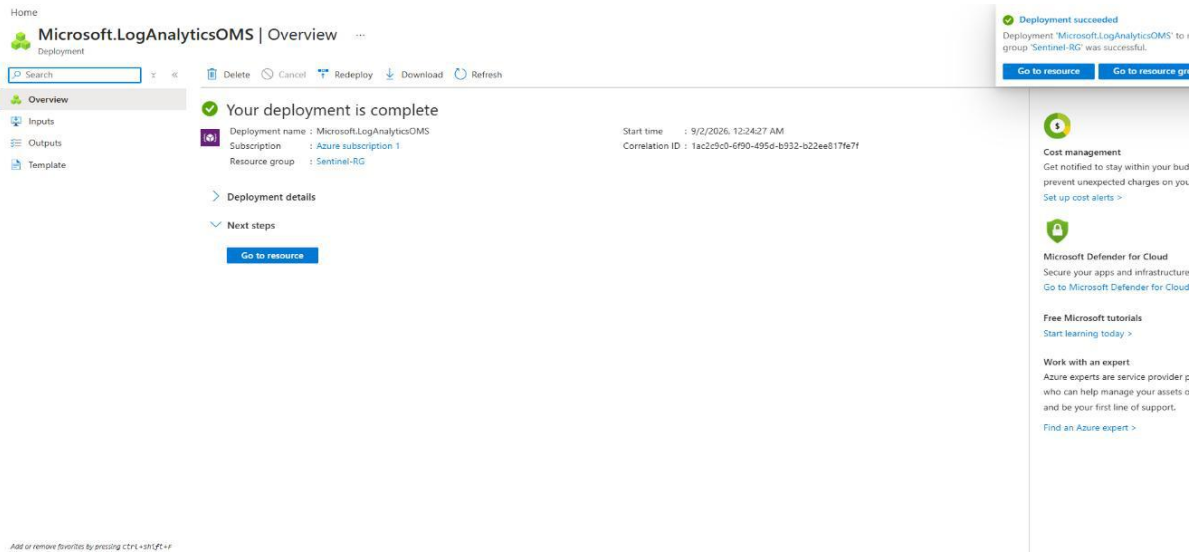
Subscription *

Resource group * [Create new](#)

Instance details

Name *

Region *



Log Analytics Workspace created successfully

Next, I configured the Azure Monitor Agent to actually get logs flowing from the VM into the workspace:

1. Went to the Content Hub in Sentinel, searched for Windows Security Events, and installed it.
2. Inside that solution, created a Data Collection Rule (DCR). This is what tells the VM which logs to collect and where to forward them, in this case, into the Log Analytics Workspace so they'd be accessible inside Sentinel.

Home

Microsoft Sentinel

Default Directory

+ Create Manage view Refresh Export to CSV Open query View incidents Add to service group Group by none

Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter

☐	Name ↑	Resource Group	Location	Subscription	Directory
☐	Honeypot-SOC-LAB	sentinel-rg	East US	Azure subscription 1	Default Directory
☐	sentinel-workspace	sentinel-rg	East US	Azure subscription 1	Default Directory

Home > Microsoft Sentinel | Content hub

Selected workspace: NonexpertsSentinel

Search

Refresh Install/Update Delete SIEM Migration Guides & Feedback

General Overview Logs Guides Search Threat management Content management Content hub Repositories Community Configuration

504 Solutions 346 Standalone contents 1 Installed 0 Updates

Didn't find what you were looking for? We're showing a limited set of results. Try refining your search for more specific results. [Learn more](#)

Search: windows security event Status: All Content type: All Support: All Provider: All Category: All Content sources: All 88

Content title	Status	Content source	Provider	Support	Category
Windows Security Events	Installed	Solution	Microsoft	Microsoft	Security
Windows Security Events via AMA	Installed	Solution	Microsoft	Microsoft	Security
Security Events via Legacy Agent	Installed	Solution	Microsoft	Microsoft	Security
Event Analyzer	Installed	Solution	Microsoft	Microsoft	Security
NNT Security event log cleared	Installed	Solution	Microsoft	Microsoft	Security
New SXS deployed via Default Domain or ...	Installed	Solution	Microsoft	Microsoft	Security
Excessive Windows Logon Failures	Installed	Solution	Microsoft	Microsoft	Security
Starting or Stopping HealthService to Avol...	Installed	Solution	Microsoft	Microsoft	Security
Gain Code Execution on ADFS Server via S...	Installed	Solution	Microsoft	Microsoft	Security

Windows Security Events

Microsoft Provider Microsoft Support 3.0.13 Version

Description

Note: Please refer to the following before installing the solution:

- Review the solution [Release Notes](#).
- There may be [known issues](#) pertaining to this Solution, please refer to them before installing.

The Windows Security Events solution for Microsoft Sentinel allows you to ingest Security events from your Windows machines using the Windows Agent into Microsoft Sentinel. This solution includes two (2) data connectors to help ingest the logs.

- Windows Security Events via AMA**: This data connector helps in ingesting Security Events logs into your Log Analytics Workspace using the new Azure Monitor Agent. Learn more about ingesting using the new Azure Monitor Agent here. **Microsoft recommends using this Data Connector.**
- Security Events via Legacy Agent**: This data connector helps in ingesting Security Events logs into your Log Analytics Workspace using the legacy Log Analytics agent.

Data Connectors: 2, Workbooks: 2, Analytic Rules: 1, Hunting Queries: 50

Manage Actions View details

Installing the Windows Security Events solution from the Content Hub

Home > Microsoft Sentinel | Content hub > Windows Security Events

Windows Security Events via AMA

Disconnected Status Microsoft Provider Last Log Received

Description

You can stream all security events from the Windows machines connected to your Microsoft Sentinel workspace using the Windows agent. This connection enables you to view dashboards, create custom alerts, and improve investigation. This gives you more insight into your organization's network and improves your security operation capabilities.

Last data received

Related content

0 Workbooks 1 Queries 20 Analytics rules templates

Data received

Go to log analytics

Prerequisites

To integrate with Windows Security Events via AMA you must:

- Workpace data sources: read and write permissions.
- To collect data from non-Azure VMs, they must have Az

Configuration

Enable data collection rule

Security Events logs are collected only from Windows ag

Refresh

Rule name Created by

No results

+ Create data collection rule

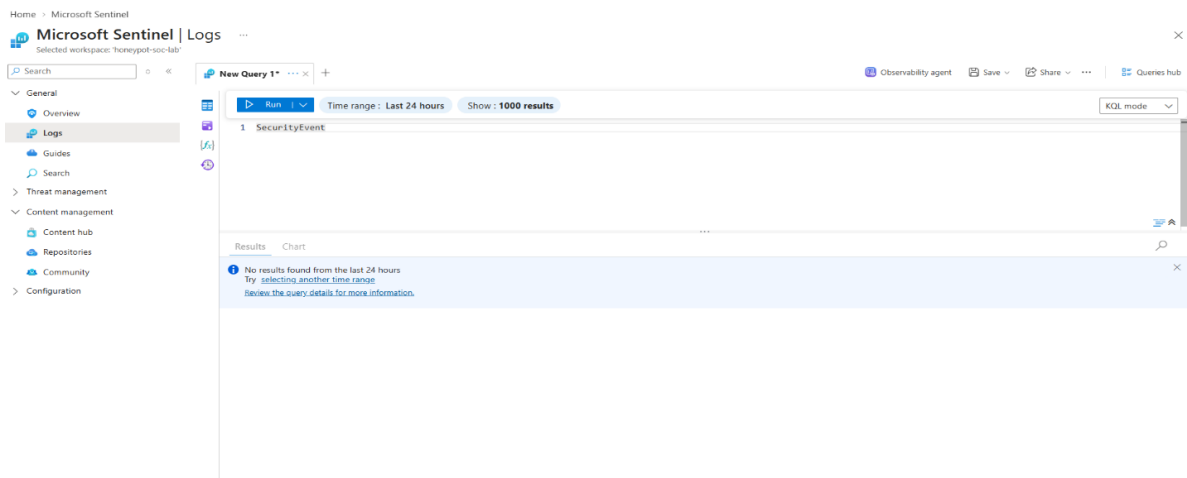
Next: Resources

Creating the Data Collection Rule and confirming the Azure Monitor Agent extension on the VM

6. Confirming Log Ingestion

Back in the Log Analytics Workspace, I ran a basic query against the SecurityEvent table to check if any logs had come in yet. Nothing showed up at first.

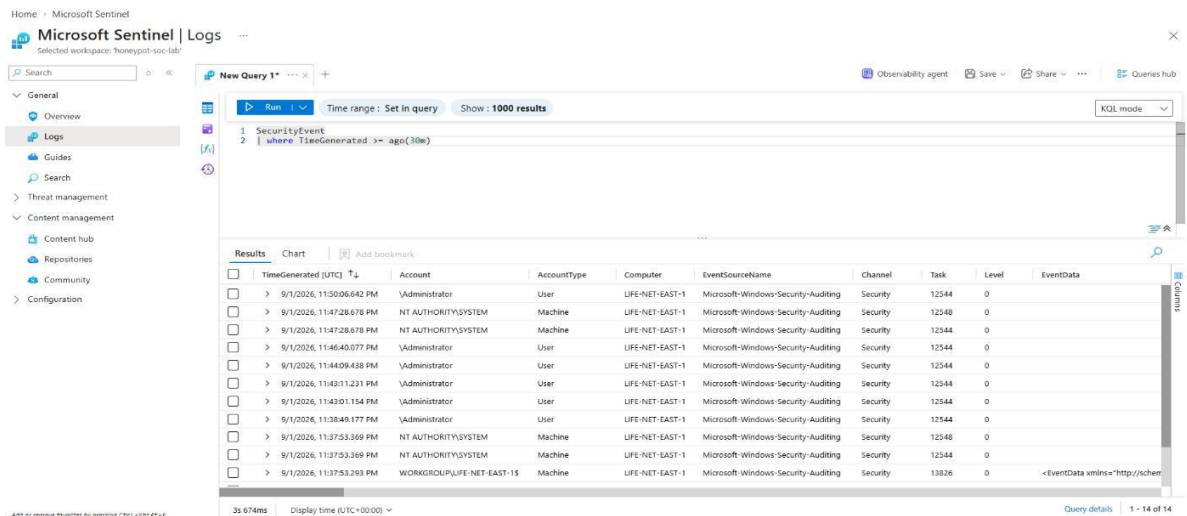
That didn't mean anything was broken. Everything had just been created moments earlier, and log ingestion takes time to start flowing. After waiting around 20 minutes and checking again, logs were showing up as expected.



No results yet, moments after setup

Lesson carried over from my other Sentinel lab: a missing log right after setup is almost always an ingestion delay, not a broken connector. Worth checking again after a wait before assuming something's misconfigured.

7. First Look at the Logs



Logs flowing in after the ingestion delay

With logs flowing, I ran a few exploratory queries to see what the honeypot was picking up, including checking the total number of failed logon attempts recorded so far.

I also pulled the IP address of one of the attackers and ran it through an IP lookup website. It resolved to Ho Chi Minh City, Vietnam. I want to be careful about what that actually proves: it tells you where the connection *appeared* to originate from, not necessarily where the attacker physically is, since VPNs and other location-masking tools are common. Digging into attacker attribution wasn't the goal of this lab, so I left that as a possible follow-up rather than chasing it here.

Home > Microsoft Sentinel | Logs

Selected workspace: honeypot-soc-lab

Search

General

- Overview
- Logs
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- Search
- Threat management
- Content management
- Content hub
- Repositories
- Community
- Configuration

New Query 1* Time range: Last 12 hours Show: 1000 results

```

1 SecurityEvent
2 | where EventID == "4625"
3 | project TimeGenerated, EventID, Account, Computer, IpAddress

```

Results Chart Add bookmark

	TimeGenerated (UTC)	EventID	Account	Computer	IpAddress
>	9/2/2026, 12:05:38.760 AM	4625	\Administrator	LIFE-NET-EAST-1	103.9.204.206
>	9/2/2026, 12:04:23.986 AM	4625	\Administrator	LIFE-NET-EAST-1	31.25.239.91
>	9/2/2026, 12:01:24.569 AM	4625	\Administrator	LIFE-NET-EAST-1	103.9.204.51
>	9/1/2026, 11:58:25.328 PM	4625	\Administrator	LIFE-NET-EAST-1	123.30.106.64
>	9/1/2026, 11:56:49.388 PM	4625	\Administrator	LIFE-NET-EAST-1	185.122.164.213
>	9/1/2026, 11:55:36.133 PM	4625	\Administrator	LIFE-NET-EAST-1	109.199.106.34
>	9/1/2026, 11:51:17.463 PM	4625	\Administrator	LIFE-NET-EAST-1	103.9.204.212
>	9/1/2026, 11:50:06.642 PM	4625	\Administrator	LIFE-NET-EAST-1	115.190.244.156
>	9/1/2026, 11:46:40.077 PM	4625	\Administrator	LIFE-NET-EAST-1	185.187.169.232
>	9/1/2026, 11:44:09.438 PM	4625	\Administrator	LIFE-NET-EAST-1	51.79.212.43
>	9/1/2026, 11:43:11.231 PM	4625	\Administrator	LIFE-NET-EAST-1	104.234.94.23

2s 658ms Display time (UTC+00:00)

Query details 1 - 12 of 13

Home > Microsoft Sentinel | Logs

Selected workspace: honeypot-soc-lab

Search

General

- Overview
- Logs
- Guides
- Search
- Threat management
- Content management
- Content hub
- Repositories
- Community
- Configuration

New Query 1* Time range: Set in query Show: 1000 results

```

1 SecurityEvent
2 | where TimeGenerated >= ago(30m)
3 | where Account == "\\Administrator"
4 | project TimeGenerated, Computer, Account, EventID, Activity, IpAddress

```

Results Chart Add bookmark

	TimeGenerated (UTC)	Computer	Account	EventID	Activity	IpAddress
>	9/1/2026, 11:58:25.328 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	123.30.106.64
>	9/1/2026, 11:56:49.388 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	185.122.164.213
>	9/1/2026, 11:55:39.155 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	109.199.106.34
>	9/1/2026, 11:51:17.463 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	103.9.204.212
>	9/1/2026, 11:50:06.642 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	115.190.244.156
>	9/1/2026, 11:46:40.077 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	185.187.169.232
>	9/1/2026, 11:43:11.231 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	51.79.212.43
>	9/1/2026, 11:43:01.154 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	104.234.94.23
>	9/1/2026, 11:43:01.154 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	185.44.65.238
>	9/1/2026, 11:38:49.177 PM	LIFE-NET-EAST-1	\Administrator	4625	4625 - An account failed to log on.	51.79.212.43

1s 7641ms Display time (UTC+00:00)

Query details 1 - 10 of 10

Failed logon events recorded by the honeypot

whatismyipaddress.com/ip/123.30.106.64

WhatIs MyIPAddress.com

Enter Keywords or IP Address... Search

ABOUT PRESS BOOK PODCAST CONTACT

MY IP IP LOOKUP HIDE MY IP VPNS TOOLS LEARN

IP Details For: 123.30.106.64

Decimal:	2065590848
Hostname:	123.30.106.64
ASN:	45899
ISP:	Vietnam Posts and Telecommunications Group
Services:	None detected
Country:	Viet Nam
State/Region:	Ho Chi Minh
City:	Ho Chi Minh City
Latitude:	10.7500 (10° 45' 0.00" N)
Longitude:	106.6667 (106° 40' 0.02" E)

ambodia ស្រុក

Cam Ranh

Ho Chi Minh City

Cần Thơ

CLICK TO CHECK BLACKLIST STATUS

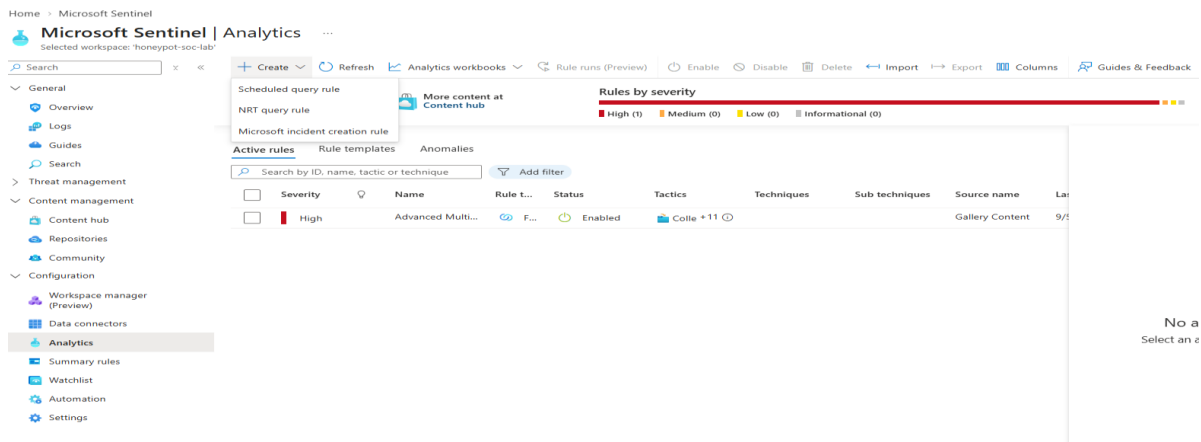
Latitude and Longitude are often near the center of population. These values are not precise enough to be used to identify a specific address, individual, or for legal purposes. IP data from IP2Location.

IP lookup showing the attacker's apparent location: Ho Chi Minh City, Vietnam

8. Building the Detection Rules

I built four detection rules, each mapped to a different MITRE ATT&CK tactic: Credential Access, Execution, Persistence, and Defense Evasion. All four were created the same way:

Microsoft Sentinel → Analytics → Create → Scheduled query rule



Creating a new scheduled query rule in Microsoft Sentinel

Rule 1: Brute Force Detection

- Name: Honeypot - Brute Force Detection
- Severity: Medium
- MITRE ATT&CK: Credential Access → Brute Force (T1110)
- Query scheduling: runs every 5 minutes, looking back over the last 2 hours
- Alert threshold: greater than 1

```
SecurityEvent
| where EventID == 4625 // failed logon
| summarize FailedAttempts = count() by IPAddress, Account
| where FailedAttempts > 5
```

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
SecurityEvent
| where EventID == 4625 // failed logon
| summarize FailedAttempts = count() by IPAddress, Account
| where FailedAttempts > 5
```

Please wait while we evaluate your query...

[View query results >](#)

Alert enhancement

> Entity mapping

> Custom details

> Alert details

Query scheduling

Run query every *

5

Minutes

Lookup data from the last *

Rule 1 query configured in the analytics rule wizard

Line by line:

- SecurityEvent: the Windows Security event table, populated by the Azure Monitor Agent.
- | where EventID == 4625: filters down to only failed logon events. This is the standard Windows event ID for a failed login.
- | summarize FailedAttempts = count() by IPAddress, Account: groups the failed attempts by the source IP address and the account being targeted, then counts how many failed attempts each IP/account pair has.
- | where FailedAttempts > 5: filters that grouped result down to pairs with more than 5 failed attempts, which is the actual brute force threshold.

Why Medium severity: a failed login attempt alone hasn't compromised anything yet, it's suspicious activity worth investigating, not confirmed impact. Brute force *attempts* sit at Medium; brute force *followed by a success* would be a stronger, High-severity signal. This is a good reminder that severity isn't just about the technique being detected, it's about impact and confidence.

Analytics rule wizard - Create a new Scheduled rule ...

Validation passed.

General Set rule logic Incident settings Automated response **Review + create**

Analytics rule details

Name	Honeypot-Brute Force Detection
Description	
MITRE ATT&CK	> Credential Access (1)
Severity	Medium
Status	Enabled

Analytics rule settings

Rule query	SecurityEvent where EventID == 4625 // failed login summarize FailedAttempts = count() by IpAddress, Account where FailedAttempts > 5
Rule frequency	Run query every 5 minutes
Rule period	Last 2 hours data
Rule start time	Automatic
Rule threshold	Trigger alert if query returns more than 1 results
Event grouping	Trigger an alert for each event
Suppression	Not configured

Entity mapping

[< Previous](#)
[Save](#)
Rule 1 review and create summary

Rule 2: Suspicious Process After Login

- Name: Honeypot - Suspicious Process After Login
- Severity: Medium
- MITRE ATT&CK: Execution → Command and Scripting Interpreter (T1059)

```

let Logons = SecurityEvent
| where EventID == 4624
| project LogonTime = TimeGenerated, Account, Computer, LogonId;
let SuspiciousProcs = SecurityEvent
| where EventID == 4688
| where NewProcessName has_any ("cmd.exe", "powershell.exe")
| project ProcTime = TimeGenerated, Account, Computer, NewProcessName;
Logons
| join kind=inner SuspiciousProcs on Account, Computer
| where ProcTime between (LogonTime .. LogonTime + 10m)
| project LogonTime, ProcTime, Account, Computer, NewProcessName

```

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
let Logons = SecurityEvent
| where EventID == 4624
| project LogonTime = TimeGenerated, Account, Computer, LogonID;
let SuspiciousProcs = SecurityEvent
| where EventID == 4688
| where NewProcessName has_any ("cmd.exe", "powershell.exe", "powershell_ise.exe")
```

[View query results >](#)

Alert enhancement

- > Entity mapping
- > Custom details
- > Alert details

Query scheduling

Run query every *

Minutes



Lookup data from the last *

Rule 2 query configured in the analytics rule wizard

Line by line:

- `let Logons = ...`: builds a temporary table of successful logon events (Event ID 4624), keeping the time, account, computer, and logon ID.
- `let SuspiciousProcs = ...`: builds a second temporary table of process creation events (Event ID 4688), filtered to just `cmd.exe` or `powershell.exe`.
- `Logons | join kind=inner SuspiciousProcs on Account, Computer`: matches logon events to process events that happened on the same account and computer.
- `| where ProcTime between (LogonTime .. LogonTime + 10m)`: narrows the match down to processes that started within 10 minutes *after* the logon. This is what turns “someone used PowerShell” into “someone used PowerShell right after logging in,” a much stronger and more specific signal than either event alone.
- `| project ...`: keeps only the fields worth seeing in the alert.

Analytics rule wizard - Create a new Scheduled rule ...



Validation passed.

General Set rule logic Incident settings Automated response **Review + create**

Analytics rule details

Name	HoneyPot-Suspicious Process After Login
Description	
MITRE ATT&CK	> Execution (T)
Severity	Medium
Status	Enabled

Analytics rule settings

Rule query	let Logons = SecurityEvent where EventID == 4624 project LogonTime = TimeGenerated, Account, Computer, LogonID; let SuspiciousProcs = SecurityEvent where EventID == 4688 where NewProcessName has_any ("cmd.exe", "powershell.exe", "powershell.exe") project ProcTime = TimeGenerated, Account, Computer, NewProcessName, CommandLine, SubjectLogonID; Logons join kind=inner SuspiciousProcs on Account, Computer where ProcTime between (LogonTime - LogonTime + 10m) project LogonTime, ProcTime, Account, Computer, NewProcessName, CommandLine
Rule frequency	Run query every 5 minutes
Rule period	Last 5 hours data
Rule start time	Automatic
Rule threshold	Trigger alert if query returns more than 0 results
Event grouping	Trigger an alert for each event
Suppression	Not configured

Rule 2 review and create summary

Rule 3: New Local User Account Created

- Name: HoneyPot - New Local User Account Created
- Severity: Medium
- MITRE ATT&CK: Persistence → Create Account (T1136), sub-technique T1136.001 (Local Account)
- Alert threshold: greater than 0

SecurityEvent

| where EventID == 4720

| project TimeGenerated, Computer, NewAccount = TargetUserName, CreatedBy = SubjectUserName

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
SecurityEvent
| where EventID == 4720
| project TimeGenerated, Computer, NewAccount = TargetUserName, CreatedBy = SubjectUserName
```

☐ Please wait while we evaluate your query...[View query results >](#)

Alert enhancement

- > Entity mapping
- > Custom details
- > Alert details

Query scheduling

Run query every *

5 Minutes

Rule 3 query configured in the analytics rule wizard

Line by line:

- | where EventID == 4720: filters to “a user account was created” events, the Windows event ID for local account creation.
- | project TimeGenerated, Computer, NewAccount = TargetUserName, CreatedBy = SubjectUserName: renames the raw fields to something readable, the account that got created, and who created it.

HOME / MICROSOFT DEFENDER | ANALYTICS

Analytics rule wizard - Create a new Scheduled rule ...

✓ Validation passed.

General Set rule logic Incident settings Automated response Review + create

Analytics rule details

Name	Honeypot-New Local User Account Created
Description	
MITRE ATT&CK	> Persistence (2)
Severity	Medium
Status	Enabled

Analytics rule settings

Rule query	SecurityEvent where EventID == 4720 project TimeGenerated, Computer, NewAccount = TargetUserName, CreatedBy = SubjectUserName
Rule frequency	Run query every 5 minutes
Rule period	Last 10 minutes data
Rule start time	Automatic
Rule threshold	Trigger alert if query returns more than 0 results
Event grouping	Trigger an alert for each event
Suppression	Not configured

Entity mapping

Rule 3 review and create summary

Why Medium severity: creating a local account isn't inherently malicious on its own (admins do it too), but on a honeypot with zero legitimate admin activity, any new account is automatically worth flagging.

Unlike the brute force rule, which needs a numeric threshold to separate normal typos from an actual attack, this rule fires on any single match. Not every detection needs a count-based threshold, some events are rare and specific enough that one occurrence is meaningful on its own.

Rule 4: Security Event Log Cleared

- Name: Honeypot - Security Event Log Cleared
- Severity: High
- MITRE ATT&CK: Defense Evasion → Indicator Removal (T1070), sub-technique T1070.001 (Clear Windows Event Logs)
- Alert threshold: greater than 0

```
SecurityEvent
| where EventID == 1102
```

```
| project TimeGenerated, Computer, ClearedBy = SubjectUserName
```

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
SecurityEvent  
| where EventID == 1102  
| project TimeGenerated, Computer, ClearedBy = SubjectUserName
```

Please wait while we evaluate your query...

[View query results >](#)

Alert enhancement

> Entity mapping

> Custom details

> Alert details

Query scheduling

Run query every *

5

Minutes

Lookup data from the last *

Rule 4 query configured in the analytics rule wizard

Line by line:

- | where EventID == 1102: filters to the specific event logged when the Windows Security audit log itself is cleared.
- | project TimeGenerated, Computer, ClearedBy = SubjectUserName: keeps the time, machine, and who triggered it.

Home > Microsoft Sentinel | Analytics

Analytics rule wizard - Create a new Scheduled rule

Validation passed.

General Set rule logic Incident settings Automated response Review + create

Analytics rule details

Name Honeypot-Security Event Log Cleared

Description

MITRE ATT&CK > Defense Evasion (2)

Severity

High

Status

Enabled

Analytics rule settings

Rule query SecurityEvent | where EventID == 1102 | project TimeGenerated, Computer, ClearedBy = SubjectUserName

Rule frequency

Run query every 5 minutes

Rule period

Last 10 minutes data

Rule start time

Automatic

Rule threshold

Trigger alert if query returns more than 0 results

Event grouping

Trigger an alert for each event

Suppression

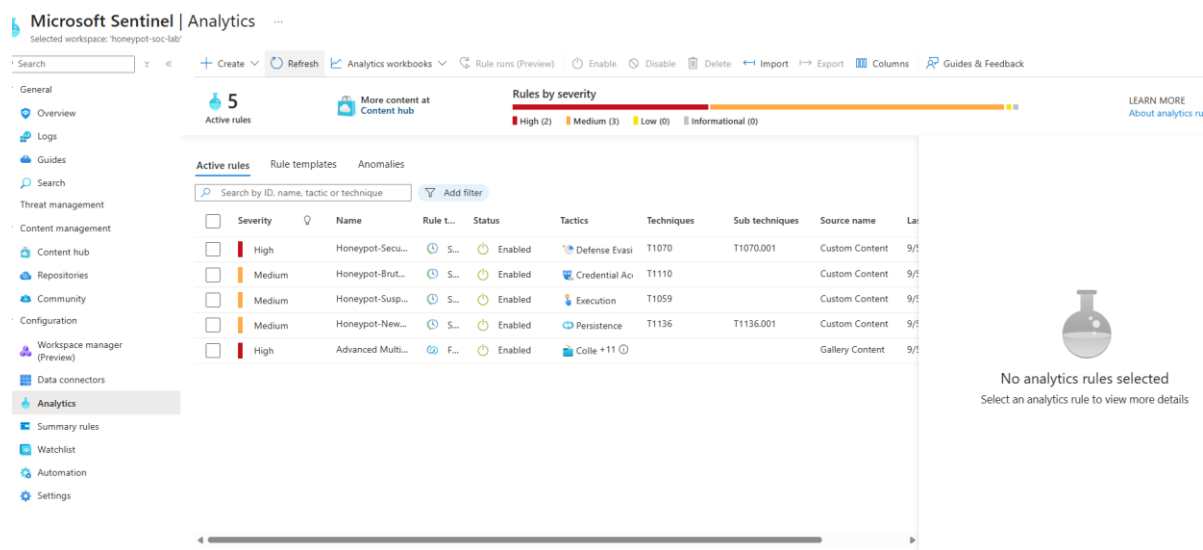
Not configured

Entity mapping

Rule 4 review and create summary

Why High severity: clearing the event log is almost never something a legitimate process on a honeypot would do. It's a classic sign that someone is actively trying to cover their tracks after doing something else first, which makes it a stronger, more confident signal than the other three rules.

A limitation worth being honest about: if an attacker successfully clears the log before this rule's next scheduled run, the very event that would have triggered the alert can get wiped along with everything else. That's a real gap in log-based detection: detecting a technique isn't the same as being resilient against it. A more mature setup would forward logs somewhere tamper-resistant, in closer to real time, which is a reasonable next step for this lab.



All four detection rules active in Microsoft Sentinel, alongside the built-in Advanced Multistage rule

9. Mapping Attackers to Real-World Locations

Knowing an attacker's IP address is useful, but seeing *where* attacks are coming from on a map makes the pattern much easier to understand at a glance. To do that, I needed to resolve IP addresses to geographic locations.

Setting up the geolocation data

I downloaded a geolocation dataset that maps IP address ranges to latitude, longitude, city, and country, and uploaded it into Sentinel as a watchlist:

```
Sentinel → Configuration → Watchlists → Create new watchlist
→ filled in name and alias
→ uploaded the file
→ Review
→ Create
```

Home > Microsoft Sentinel | Watchlist

Watchlist wizard

General **Source** Review + create

Source type * Local file

File type * CSV file with a header (.csv)

Number of lines before row with headings * 0

Upload file * geoiip-summarized.csv

Drag and drop the files or [Browse for files](#)

SearchKey * network

[Reset](#)

File preview | First 50 rows and first 5 columns

network	latitude	longitude	cityname	countryname
1.0.0.0/16	-33.494	143.2104		Australia
1.1.0.0/16	17.8148	103.3386	Ban Chan	Thailand
1.2.0.0/16	13.8667	100.1917	Nakhon Pathom	Thailand
1.3.0.0/16	13.8679	100.1891	Nakhon Pathom	Thailand
1.4.0.0/16	13.6687	100.579	Bangkok	Thailand
1.5.0.0/16	13.6659	100.5882	Bangkok	Thailand
1.6.0.0/16	12.9634	77.5855	Bengaluru	India
1.7.0.0/16	12.9691	77.5902	Bengaluru	India
1.8.0.0/16	12.9557	77.5843	Bengaluru	India
1.9.0.0/16	3.1539	101.7448	Ampang	Malaysia

Uploading the geolocation dataset as a Sentinel watchlist

Home > Microsoft Sentinel | Watchlist

Selected workspace: Honeypot-soc-lab

Search

Refresh New Delete Update watchlist Columns Guides & Feedback

General Overview Logs Guides Search Threat management Content management Configuration

Workspace manager (Preview) Data connectors Analytics Summary rules **Watchlist** Automation Settings

1 Watchlists **53K** Watchlist items

My Watchlists Templates (Preview)

Search by name, alias and description Add

Name	Alias	Source	Create...	Last u...
geoiip	geoiip	geoiip-summarized.csv	9/2/2026	9/2/2026

geoiip

Microsoft Provider 53K Rows 9/2/2026, 10:35... Created time

Description

Source: geoiip-summarized.csv

Created by

Last updated: 9/2/2026, 10:35:24 PM

SearchKey: network

Status (Preview): Succeeded

[View in logs](#) [Update watchlist](#)

< Previous Page 1 of 1 Next > Showing 1 to 1 of 1 results.

The geoiip watchlist, holding 53K rows

The idea is that for any attacker IP address, Sentinel can check which range it falls into in this dataset and pull out the matching location.

Confirming the lookup worked

```
let GeoIPDB_FULL = _GetWatchlist("geoiip");
SecurityEvent
| where EventID == 4625
| order by TimeGenerated desc
| evaluate ipv4_lookup(GeoIPDB_FULL, IPAddress, network)
| project TimeGenerated, Computer, AttackerIp = IPAddress, cityname, countryname
```

Line by line:

- `let GeoIPDB_FULL = _GetWatchlist("geoip")`: loads the uploaded geolocation watchlist into a variable so it can be referenced in the query.
- `| where EventID == 4625`: narrows to failed logon events, same as the brute force rule.
- `| order by TimeGenerated desc`: sorts the results with the most recent event first.
- `| evaluate ipv4_lookup(GeoIPDB_FULL, IPAddress, network)`: this is the actual geolocation step. It checks the attacker's IP address against the IP ranges in the watchlist and attaches the matching location data (city, country, coordinates) to each row.
- `| project ...`: keeps just the time, machine, attacker IP, and resolved city/country.

Microsoft Sentinel | Logs

Time range: Last 12 hours | Show: 1000 results

```

1 let GeoIPDB_FULL = _GetWatchlist("geoip");
2 let WindowEvents = SecurityEvent
3 | where EventID == "4625"
4 | evaluate ipv4_lookup(GeoIPDB_FULL, IPAddress, network);
5 WindowEvents
6 | project TimeGenerated, Computer, AttackerIP=IPAddress, cityname, countryname, latitude, longitude

```

TimeGenerated [UTC]	Computer	AttackerIP	cityname	countryname	latitude	longitude
9/2/2026, 10:24:35.557 PM	LIFE-NET-EAST-1	192.220.24.193	Orlando	United States	28.5208	-81.3544
9/2/2026, 10:19:27.741 PM	LIFE-NET-EAST-1	47.181.122.192	Flower Mound	United States	33.024	-97.1187
9/2/2026, 10:15:15.13 PM	LIFE-NET-EAST-1	20.48.238.59	Singapore	Singapore	1.2867	103.8496
9/2/2026, 10:13:46.403 PM	LIFE-NET-EAST-1	47.181.122.192	Flower Mound	United States	33.024	-97.1187
9/2/2026, 10:13:36.630 PM	LIFE-NET-EAST-1	109.123.238.248	Sint-Katelijn...	Belgium	51.0677	4.5319
9/2/2026, 10:11:44.033 PM	LIFE-NET-EAST-1	103.9.204.47	Sydney	Australia	-33.8715	151.2006
9/2/2026, 10:09:40.471 PM	LIFE-NET-EAST-1	156.67.31.88	Cape Town	South Africa	-33.914	18.4129
9/2/2026, 10:08:19.896 PM	LIFE-NET-EAST-1	31.25.239.91	Newton Aycliffe	United Kingdom	54.6	-1.55
9/2/2026, 10:08:19.896 PM	LIFE-NET-EAST-1	31.25.239.91	Newton Aycliffe	United Kingdom	54.6063	-1.5541
9/2/2026, 10:08:19.896 PM	LIFE-NET-EAST-1	31.25.239.91	Olbia	Italy	40.9302	9.5034
9/2/2026, 10:07:49.734 PM	LIFE-NET-EAST-1	47.181.122.192	Flower Mound	United States	33.024	-97.1187
9/2/2026, 10:06:07.878 PM	LIFE-NET-EAST-1	51.79.212.43	London	United Kingdom	51.5095	-0.0655

Geolocation lookup query and results

Microsoft Sentinel | Logs

Time range: Last 12 hours | Show: 1000 results

```

1 SecurityEvent
2 | where EventID == "4625"
3 | project TimeGenerated, EventID, Account, Computer, IPAddress

```

TimeGenerated [UTC]	EventID	Account	Computer	IPAddress
9/2/2026, 12:05:38.768 AM	4625	VAdministrator	LIFE-NET-EAST-1	103.9.204.206
9/2/2026, 12:04:23.986 AM	4625	VAdministrator	LIFE-NET-EAST-1	31.25.239.91
9/2/2026, 12:01:24.569 AM	4625	VAdministrator	LIFE-NET-EAST-1	103.9.204.51
9/2/2026, 11:58:25.328 PM	4625	VAdministrator	LIFE-NET-EAST-1	123.30.106.64
9/1/2026, 11:56:49.388 PM	4625	VAdministrator	LIFE-NET-EAST-1	185.122.164.213
9/1/2026, 11:55:39.133 PM	4625	VAdministrator	LIFE-NET-EAST-1	109.199.106.34
9/1/2026, 11:51:17.463 PM	4625	VAdministrator	LIFE-NET-EAST-1	103.9.204.212
9/1/2026, 11:50:06.642 PM	4625	VAdministrator	LIFE-NET-EAST-1	115.190.244.156
9/1/2026, 11:46:40.077 PM	4625	VAdministrator	LIFE-NET-EAST-1	185.187.169.232
9/1/2026, 11:44:09.458 PM	4625	VAdministrator	LIFE-NET-EAST-1	51.79.212.43
9/1/2026, 11:43:11.231 PM	4625	VAdministrator	LIFE-NET-EAST-1	104.234.94.23

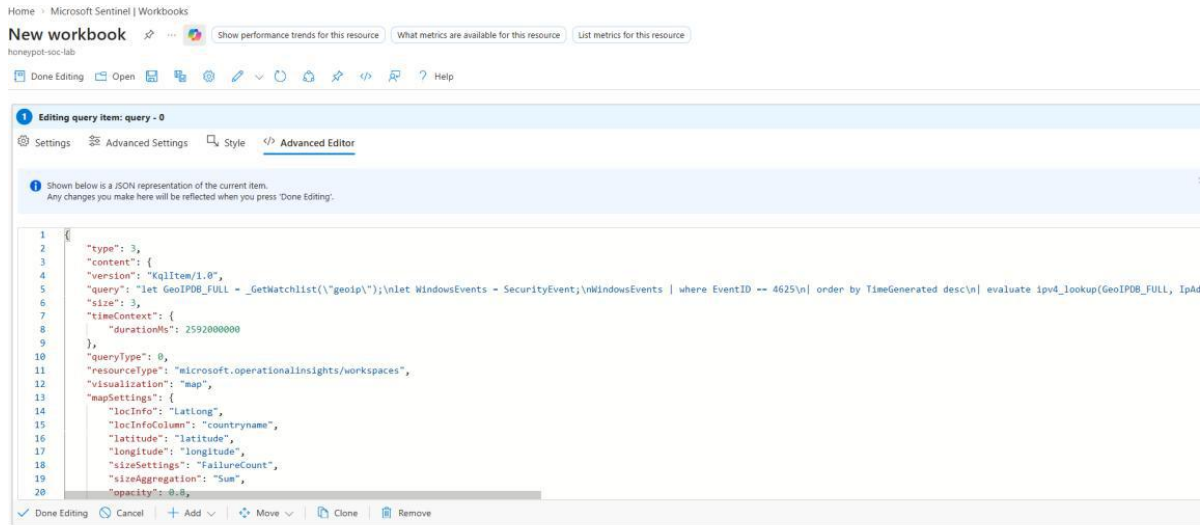
Failed logon events with attacker IP addresses

Before integrating the watchlist, the raw security event logs had no location information at all, just IP addresses. After this query, each event carried city and country data alongside it.

10. Building the Attacker Map

With location data resolvable, the last step was visualizing it. I built this using a Sentinel workbook:

Sentinel → Workbooks → Add workbook → Edit
(removed the default template elements)
→ Add query → Advanced Editor
→ pasted in the geoip-based query as JSON
→ saved



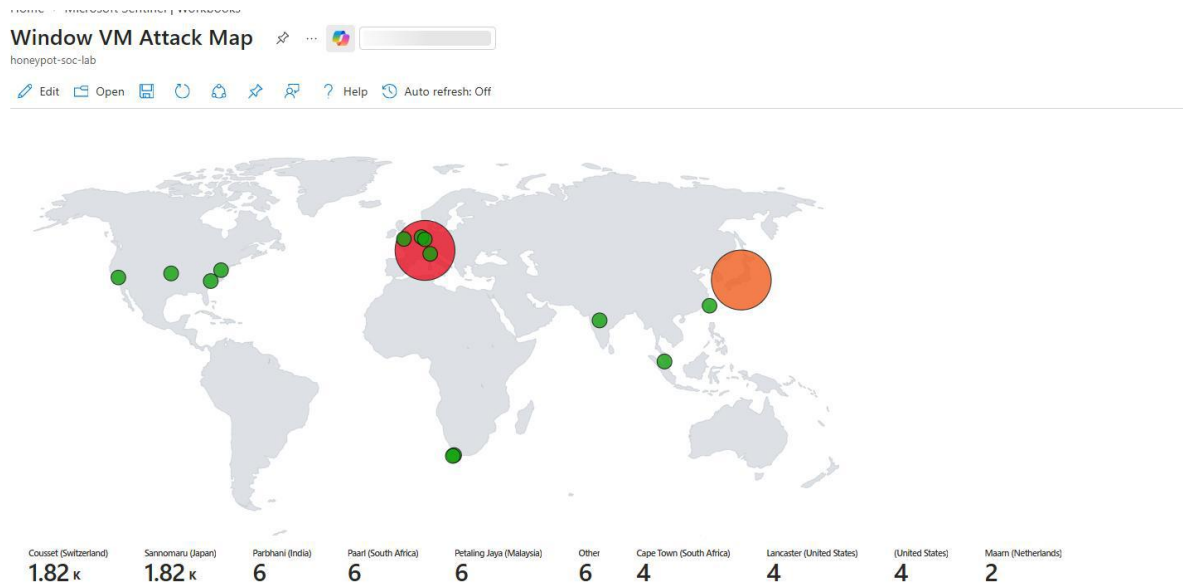
Building the map query in the workbook's Advanced Editor

This turns raw log rows into a live map showing every attacker's approximate location, updating as new failed logons and geolocation matches come in.

11. Results

I left the VM running for 24 hours to see what it would attract. In that window:

- The map showed a real spread of attacker locations from around the world hitting the VM.
- The brute force detection rule fired, and an incident was created in Sentinel from it, confirming the detection logic actually worked against real attack traffic, not just test data.



Live attacker map after 24 hours — attacks from Vietnam, the US, South Africa, Malaysia, India, the UK, Italy, and more

Home > Microsoft Sentinel

Microsoft Sentinel | Incidents

Selected workspace: 'honeypot-soc-lab'

Search by ID, title, tags, owner or product

Severity: All Status: 2 selected Incident Provider name: All

Auto-refresh incidents

Severity	Incident number	Title	Alerts	Incident provider name	Alert product name	Created time
Medium	1	Honeypot-Brute For...	2	Microsoft Defender ...	Microsoft Sentinel	09/05/26, 07:02 PM

Open incidents by severity: High (0), Medium (1), Low (0), Informational (0)

Honeypot-Brute Force Detection
Incident number 1
Investigate in Microsoft Defender XDR

Unassigned Owner New Status Medium Severity

Alert product names: Microsoft Sentinel

Evidence: 2 Events, 2 Alerts, 0 Bookmarks

Last update time: 09/05/26, 07:02 PM
Creation time: 09/05/26, 07:02 PM

Entities (0)

Tactics and techniques: Credential Access (1)

The investigation graph requires that your incident includes entities (for example: user, host, IP, etc.). Use the entity mapping option when defining your alerts. [Learn more...](#)

View full details Actions

The brute force detection rule fired, generating a real incident

Home > Microsoft Sentinel | Incidents

Honeypot-Brute Force Detection

Incident number 1

Refresh Logs Tasks Activity log

This is the new, improved incident page - **Now generally available**. You can use the toggle to switch back.

Medium Severity New Status Unassigned Owner

Investigate in Microsoft Defender XDR

Workspace name
honeypot-soc-lab

Description
--

Alert product names
• Microsoft Sentinel

Evidence
2 Events 2 Alerts 0 Bookmarks

Last update time
9/5/2026, 7:02:40 PM

Creation time
9/5/2026, 7:02:39 PM

Entities (0)
-

Tactics and techniques

The investigation graph requires that your incident includes entities (for example: user, host, IP, etc.). Use the entity mapping option when defining your alerts. [Learn more >](#)

Overview Entities

Incident timeline

Sep 5 17:57:19 Honeypot-Brute Force Detection Me... Detected by Micros... Ta...

Sep 5 17:57:19 Honeypot-Brute Force Detection Me... Detected by Micros... Ta...

Entities

No Entities
No entities found

Similar incidents

No similar incidents to display

Incident detail view for the Honeypot - Brute Force Detection rule

I didn't investigate that incident in depth here, since deep incident investigation was outside the scope of this particular lab (that's more the shape of my Macro-ni-style investigation work). But having a real, organically-triggered incident is solid proof that the detection pipeline works end to end.

12. What This Project Taught Me

- **Exposure has consequences you can measure.** It's one thing to know a system on the open internet gets attacked; it's another to watch it happen in real time on a map within hours of opening the firewall.
- **A honeypot is only as useful as its logging.** None of the detection rules mean anything without the Log Analytics Workspace, Azure Monitor Agent, and Data Collection Rule correctly wired up first. Getting the pipeline right came before writing a single detection query.
- **Severity should reflect confidence, not just technique.** A failed login is Medium. A cleared log is High. The technique matters, but so does how strongly the behavior implies something bad actually happened.
- **Detections have blind spots, and that's worth documenting, not hiding.** The event-log-cleared rule can be evaded by an attacker who clears logs fast enough. Writing that down honestly is more valuable than pretending the rule is airtight.
- **Geolocation data adds context, not proof.** An IP resolving to a city doesn't confirm where an attacker physically is. It's a useful signal, not a conclusion.

13. Summary

I built a honeypot in Azure by creating a resource group, virtual network, and VM, then deliberately exposed it by opening the network security group to all inbound traffic and disabling the VM's internal firewall. I set up a Log Analytics Workspace connected to Microsoft Sentinel, and used the Azure Monitor Agent with a Data Collection Rule to forward Windows security logs into that workspace. After confirming logs were flowing, I built four detection rules covering brute force login attempts, suspicious processes launched right after login, new local user account creation, and event log clearing, mapped respectively to the Credential Access, Execution, Persistence, and Defense Evasion MITRE ATT&CK tactics. I then integrated a geolocation watchlist to resolve attacker IP addresses to real-world locations and built a Sentinel workbook to visualize those attacks on a live map. After leaving the VM exposed for 24 hours, real attack traffic from around the world triggered the brute force detection rule and generated an actual incident, confirming the detection pipeline worked against genuine, unsimulated attacks.

14. Resources

- [Microsoft Sentinel documentation](#)
- [Microsoft Azure Network Security Group documentation](#)